

BME1307 Project 2026 Spring

Report due: 1:00PM, 6/25/2026

General Instructions:

Each group (3 persons) will have 20 minutes for presentation (16 minutes to present and 4 minutes Q/A section) on 6.11 and 6.16 class. The presentation order is determined by randomized sampling. Please control your presentation time (Overtime will result in point deduction).

In addition, each group will need to submit a written report of the project (group members will share the same grade, only one report per group).

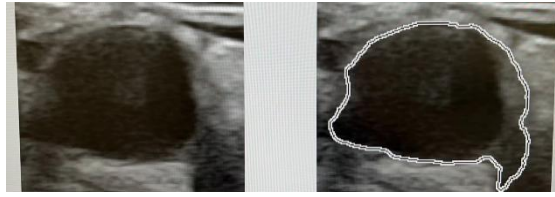
Part 1: Ultrasound image acquisition and segmentation (40 points)

(a) Acquire the real ultrasound image (20 points)

1. We will use portable ultrasound machine to capture the real ultrasound image from your neck (left or right is OK). In the captured image, thyroid region and carotid artery should be visible. Trachea is not necessary, however, it can help you to identify the location. Hint: Please check the user manual in BB before doing this.
2. Once you find the right location, please capture the images at different Gains (增益), dynamic ranges (动态范围) for both ultrasound B-mode image and Color Doppler image. You need to analyze these two parameters in post-processing.
3. Save all images for post-processing.
4. Remember to record your settings such as acquired depth, and record the size or diameter of your carotid artery (the default tool provided by the software).

(b) Image segmentation (20 points)

Healthy subjects won't have abnormal thyroid nodes; therefore, we will do image segmentation for B-mode and/or color Doppler carotid artery (see an example attached).



1. Please implement any techniques you learned in class or from other places to segment and quantify the artery. (should you crop the image if your segmentation algorithm is not robust)
2. Based on the segmentation results, please measure the diameter, the area, the similarity to the circle, .etc. And then compare with literatures. For instance, the common carotid artery lumen diameter ranges from 4.3 mm to 7.7 mm, are your measurement in a reasonable range? (Hint: it is necessary to find out the real distance equivalent to a single pixel during your data acquisition procedure, DO NOT FORGET IT in your in-class scanning)
3. Can we apply BUSAT toolbox (mentioned below) to segment carotid artery? If yes, do it and explain why. If not, please state the reason.

Part 2: Ultrasound image classification (30 points)

In this part, you can directly use “autosegment” function in BUSAT toolbox to help you segment the breast ultrasound image. The MATLAB toolbox can be downloaded <https://www.tamps.cinvestav.mx/~wgomez/downloads.html>.

Feature extraction and selection are important steps for following classification task. What you need to do is to select/quantify the features based on segmentation results, followed by lesion classification task. A total of 120 ultrasound samples and associate labels is given at <https://github.com/Qian-IMMULab/Breast-ultrasound-samples> .

Answer the following questions:

1. Implement at least **three** classification approaches, present your results using cross-validation strategy. Evaluation matrices include accuracy, sensitivity, specificity, and ROC curve with AUC value.
2. Does the number of selected features affect the performance of classification? Please justify your explanation.
3. Segmentation error is evitable in practice. Do you think it can affect the

classification results? Please justify your answer using examples.

Part 3: Foundation model-driven analysis (20 points)

In recent years, medical image analysis has undergone rapid progress, particularly with the transformative impact of foundation-model-driven approaches. In this section, please investigate generalist foundation models and ultrasound-specific foundation models, evaluate their performance on the Part 1 segmentation task and the Part 2 classification task, and provide a detailed analysis of their strengths and limitations.

Hint: To reduce computational demand, you may directly evaluate the models without training or fine-tuning.

Part 4: Your thoughts (10 points)

Open mind questions (Not limited to ultrasound):

1. Share the ideas or methods you think can improve the medical image processing. Either segmentation, classification or end-to-end is OK. You need to be specific.
2. The future direction of medical imaging processing? Any concerns for medical applications.